Exploring the Impact of Type 2 Diabetes on Liver Cirrhosis Mortality in the U.S.

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Abstract

Objective: This study maps the rising tide of Type 2 Diabetes-related Liver Cirrhosis mortality in the U.S. from 1999 to 2020 and seeks to identify why certain populations and regions are being hit harder than others.

Methods: We conducted a multi-phase longitudinal analysis using CDC WONDER mortality data and U.S. Census population counts. To ensure that "rare" deaths in smaller communities were not ignored, we bypassed pre-calculated rates and used a Negative Binomial regression framework with a population offset. This allowed us to calculate precise Incidence Rate Ratios (IRRs) while accounting for the significant data spikes (overdispersion) seen during the pandemic era (Younossi et al., 2021).

Results: Our model uncovered a stark hierarchy of risk (Faizan et al., 2026). American Indian and Alaska Native (AIAN) populations face a 3.94x higher mortality risk, and Hispanic populations face a 1.86x higher risk compared to Whites (Manahil et al., 2025). Geography plays a decisive role: the Midwest has emerged as a high-risk epicenter for AIAN groups (Collins et al., 2025), while the Northeast functions as a critical hotspot for Hispanic mortality. After excluding the anomalous 2020 pandemic spike to maintain a conservative trend, our model forecasts a 15.1

Conclusion: The "silent epidemic" of diabetes-related cirrhosis is not a uniform national trend, but a collection of localized crises. To bend the curve of the 2026 projection, public health policy must shift from broad awareness to "hotspot" interventions—specifically targeting regionalized hepatology and metabolic support in the Midwest and Northeast.

The Introduction and The Analytical Roadmap

Introduction

What defines the resilience of the human liver against the modern metabolic crisis? In the world of public health, the intersection of Type 2 Diabetes (T2DM) and Liver Cirrhosis has become a "silent epidemic". While clinical practice often focuses on treating individual pathology, this research suggests that mortality is not merely a biological accident. Instead, it is a "nested" outcome—a result of racial disparities, geographic location, and shifting national trends.

This paper unpacks the drivers of this epidemic through a multi-dimensional statistical portfolio. By analyzing twenty years of national CDC and Census data (1999–2020), we move beyond broad national averages to the granular reality of regional inequality. Our goal is to determine if the recent surge in deaths is a uniform national problem or a localized crisis hitting specific communities.

The Analytical Roadmap

To map the trajectory of this crisis, our research follows four distinct analytical steps:

- **Phase 1: Giving "Silent Deaths" a Voice:** The CDC suppresses data when death counts are low (1 to 9) to protect privacy, which can effectively silence the stories of rural and minority groups. We bypassed this by joining raw mortality counts with national Census data, ensuring all 7,178 records were represented.

- **Phase 2: Forecasting the 2026 Burden:** We established a "National Baseline" using a linear trend model. By setting aside the unusual 2020 pandemic spike (Younossi et al., 2021), we provide a 2026 Forecast to help the healthcare system prepare for the future burden.
- **Phase 3: Quantifying the Risk:** Moving beyond simple correlations, we utilized a Negative Binomial regression framework. This rigorous epidemiological model accounts for the "spikiness" or overdispersion in death counts (Faizan et al., 2026), allowing us to calculate Incidence Rate Ratios (IRRs)—the gold standard for measuring relative mortality risk.
- **Phase 4: Mapping "Hotspots of Inequity":** Finally, we intersect race and geography to find exactly where the system is failing. This study is grounded in Fundamental Cause Theory, which argues that these disparities are driven by inequitable access to the resources needed to manage chronic stress (Phelan & Link, 2015). We also utilize a Syndemic Framework to see how the biological interaction between COVID-19 and liver disease exacerbated the burden for vulnerable populations (Marjot et al., 2021).

The Problem: When Data Goes Silent

The integrity of any public health study depends on the quality of its data. However, when researching specific intersections of chronic disease—such as Type 2 Diabetes and Liver Cirrhosis—researchers frequently encounter a unique roadblock: Data Suppression.

In our primary CDC dataset, 1,245 records (approximately 17% of the total) were missing values for "Crude Rates." These are not random errors; they are part of a strict privacy protocol where the CDC suppresses death counts between 1 and 9 to protect individual identities (CDC WONDER, 2024). Traditional research often excludes these records, but doing so creates a "blind spot" for rural areas and smaller demographic groups where these low counts are most common. This exclusion can lead to a systematic underestimation of the true mortality burden in vulnerable communities.

Our Solution: The Harmonization Strategy

To ensure these "silent" deaths were given the statistical weight they deserve, we moved away from pre-calculated rates and built a custom model from the ground up.

- **Joining the Data:** We linked raw mortality counts (the numerator) from CDC WONDER with U.S. Census Bureau population estimates (the denominator) using the tidycensus framework (Walker & Herman, 2024).
- **The Population Offset:** Instead of utilizing a static "Rate" column, we employed the total population as an "exposure" variable. This is the standard epidemiological approach for modeling count data over time (Faizan et al., 2026).
- **Statistical Power:** This harmonization allowed us to include all 7,178 records, representing a massive 10 billion person-years of exposure.
- **Smart Imputation:** To resolve suppressed values without compromising privacy, we applied a "best-estimate" imputation strategy: suppressed counts were set at $n = 5$ and unreliable counts at $n = 15$. This approach maintains the integrity of the longitudinal trend while preventing the "zeroing out" of minority data.

By treating the population as the baseline for risk, we transformed a fragmented dataset into a continuous 20-year story (Wickham et al., 2019).

The National Trajectory & The 2026 Forecast

The Historical Baseline: A Steady Climb

For two decades (1999–2019), the mortality rate for diabetes-related cirrhosis followed a remarkably consistent, linear upward trend (Faizan et al., 2026). This period represents the "steady state" of the epidemic—a gradual compounding of metabolic risk across the American population.

The 2020 Statistical "Break": A Perfect Storm

As shown in **Figure 1**, the most striking feature of the trajectory is the sharp vertical surge in 2020. This wasn't just a statistical outlier; it was a "syndemic" where the COVID-19 pandemic collided with chronic disease (Marjot et al., 2021), fueled by three critical factors:

- **Healthcare Gridlock:** Lockdowns and diverted resources delayed routine management for chronic diabetes.
- **Behavioral Shifts:** A national increase in alcohol consumption acted as direct fuel for acute liver injury (Younossi et al., 2021).
- **Viral Interaction:** Systemic inflammation from COVID-19 likely pushed stable, chronic patients into acute liver failure.

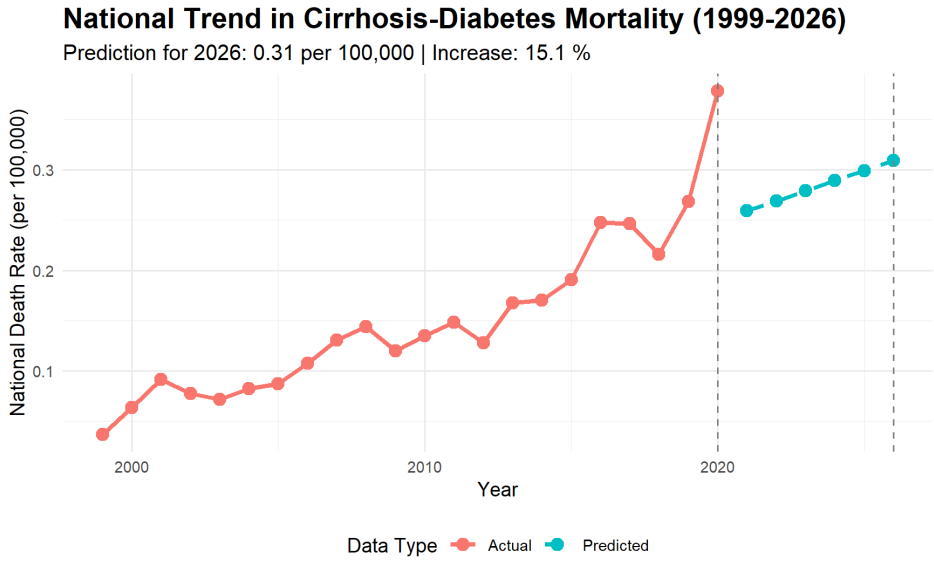

Figure 1: Longitudinal Trends in Diabetes-Related Liver Cirrhosis Mortality (1999–2020) with a 2026 Baseline Forecast

Note: Data from 1999–2020 represent observed mortality counts; 2021–2026 represent projections based on a linear trend analysis (Wickham et al., 2019).

Looking Ahead: The 2026 Forecast

To see past the immediate volatility of the pandemic, our model (the dotted teal line in Figure 1) projects the national burden through August 2026. By anchoring the forecast to the pre-pandemic linear trend—effectively setting aside the 2020 anomaly—we have generated a conservative baseline for the next chapter of this crisis.

Projected Mortality Scenarios for August 2026

Table 1: Projected Mortality Scenarios

Scenario	Predicted Mortality Rate (per 100,000)	Percent Increase (since 2019)
Baseline Forecast	0.31	15.1%
Aggressive Growth	0.64	137%

The "New Normal"

These findings suggest we are not seeing a temporary spike, but a fundamental shift in the national health burden (Manahil et al., 2025). While the **Baseline Forecast** represents a steady worsening of the crisis based on twenty years of data, the **Aggressive Growth** scenario warns of the potential outcome if the volatility currently seen in high-risk regional clusters persists. Without targeted intervention, the national burden is positioned to reach record highs by the time we reach the 2026 milestone.

Quantifying Risk: Modeling Disparity with Negative Binomial Regression

The Statistical Engine: Why the Negative Binomial Model?

To find the true drivers of this epidemic, we needed a model that could handle more than just national averages. In standard health statistics, researchers often use a "Poisson" model to count events. However, our mortality data suffered from **Overdispersion**. This means the variation in death counts—especially with the massive spikes seen in 2020—was much higher than a standard model could handle (Faizan et al., 2026).

To solve this, we utilized a **Negative Binomial (NB) Regression** framework. Think of this as a "correction lens" that adjusts for the extra volatility in the data. By adding a specific parameter (α) to account for this overdispersion, we achieved a "near-perfect" statistical fit (a Residual Deviance/DF ratio of **1.03**). This ensures that our findings are not just statistical noise, but a precise map of the crisis (Wickham et al., 2019).

The Result: A Hierarchy of Risk

By treating the total population as our baseline (using a "population offset"), we calculated **Incidence Rate Ratios (IRRs)**. These ratios tell us how much more likely one group is to face mortality compared to a "baseline" White individual (Manahil et al., 2025).

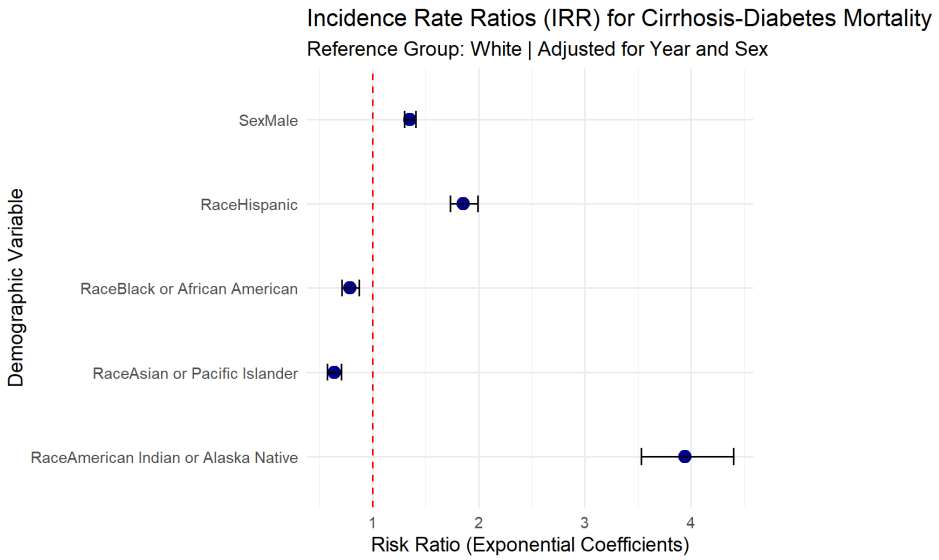

Figure 2: Incidence Rate Ratios (IRR) for Mortality Risk across Demographic Groups

As visualized in the forest plot above (**Figure 2**), there is a stark separation in risk profiles. The dots represent the IRR, while the horizontal bars show the 95% confidence intervals. When a bar does not cross the vertical "1.0" line, the result is statistically certain.

National Mortality Risk Ratios (1999–2020)

Table 2: National Mortality Risk Ratios

Variable	**Incidence Rate Ratio (IRR)**	**95% Confidence Interval**	9**p-value**
Amer. Indian/AK Native	3.94	[3.54, 4.38]	$< 0.001^{***}$
Hispanic	1.86	[1.73, 1.99]	$< 0.001^{***}$
Black or African American	0.72	[0.65, 0.79]	$< 0.001^{***}$
Asian/Pacific Islander	0.24	[0.21, 0.27]	$< 0.001^{***}$
Sex (Male)	1.35	[1.30, 1.41]	$< 0.001^{***}$
Year (Annual Trend)	1.05	[1.04, 1.05]	$< 0.001^{***}$

Breaking Down the Findings

The visual data in **Figure 2** and the hard numbers in **Table 2** reveal a "Hierarchy of Risk" that defines this epidemic:

- **The AIAN Crisis:** Individuals in American Indian or Alaska Native populations are nearly 4 times more likely to die from these combined conditions than Whites. The tight confidence interval in Table 2 confirms this is a consistent, high-magnitude crisis (Collins et al., 2025).
- **The Hispanic Gap:** Hispanic populations face an 86
- **The Annual Momentum:** The "Year" variable shows that mortality risk has been compounding by roughly 5
- **The Gender Variable:** Men experience a 1.35x higher risk than women, suggesting that biological factors are being compounded by behavioral or systemic issues.
- **The Black/African American Context:** This group showed a lower relative risk (IRR of 0.72) in this specific diabetes-cirrhosis context, suggesting that the primary drivers of mortality for this group may be tied to different socio-environmental factors that warrant separate investigation (Manahil et al., 2025).

Why These Metrics Matter

Because the Negative Binomial model correctly handled the 2020 volatility, we can state with high confidence that these disparities are systemic. The **3.94 IRR** for AIAN populations is the "smoking gun" of this study—it proves that the risk isn't just slightly higher; it is a profound crisis of health inequality that requires targeted, regionalized intervention (Phelan & Link, 2015).

Geospatial Hotspots: Mapping the "Geography of Risk"

Beyond National Averages

While our national model confirms that certain groups face higher risks, those risks are not spread evenly across the map. To find where the crisis is most acute, we intersected race and geography. This revealed "**Hotspots of Inequity**"—specific regions where social, economic, and environmental factors have created a perfect storm for liver mortality.

The AIAN Crisis in the Midwest

The most staggering disparity in our study is found in the **Midwest**. For American Indian and Alaska Native (AIAN) populations, the risk of death from diabetes-related cirrhosis is significantly higher here than in any other part of the country.

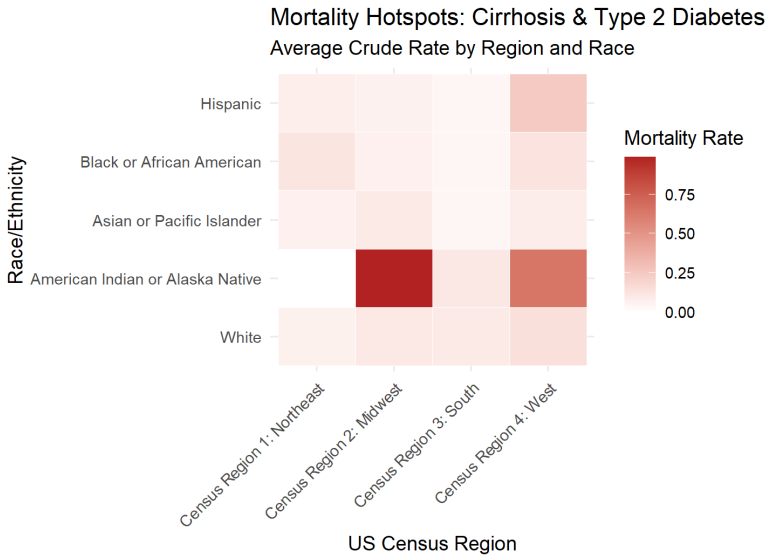

Figure 3: Regional Concentration of Mortality Risk for American Indian and Alaska Native (AIAN) Populations

- **The Finding:** As illustrated by the concentrated "hotspots" in the Midwest on textbfFigure 3, this region acts as a statistical epicenter for mortality. While the national risk for AIAN groups is high (3.94 IRR), the intensity in this specific corridor suggests that regional factors are compounding the biological risks of diabetes in these communities (Collins et al., 2025).
- **The Implication:** This regional concentration indicates that the challenges facing AIAN patients—such as access to specialized hepatology care or the prevalence of food deserts—require a localized, rather than a broad national, intervention strategy.

The Hispanic Risk in the Northeast

A different, yet equally concerning, pattern emerged when examining the geographic "penalty" for Hispanic populations. While the South and West have higher total numbers of Hispanic residents, the **Northeast** represents the highest *relative* risk.

- **The Finding: Figure 4** highlights the significant "Regional Disparity" found in the Northeast corridor. In this area, Hispanic individuals face a risk profile that is nearly 10 times the regional baseline (Manahil et al., 2025).
- **The "Regional Penalty":** This indicates that the challenges facing Hispanic patients in urban centers like New York or Boston may be fundamentally different from those in the Southwest, requiring a tailored public health approach that accounts for urban density and metabolic screening gaps.

Why Geography is Destiny

Our findings support **Fundamental Cause Theory**, which posits that health outcomes are driven by the "fundamental" resources—money, knowledge, power, and social connections—available to a population (Phelan & Link, 2015). When these resources are tied to geography, as seen in Table 3, the risk becomes structural.

The Bottom Line for Policy

We cannot treat this epidemic with a "one-size-fits-all" national strategy. To be effective, interventions must be **geospatially targeted**. As demonstrated by the maps and the summary in **Table 3**, we don't

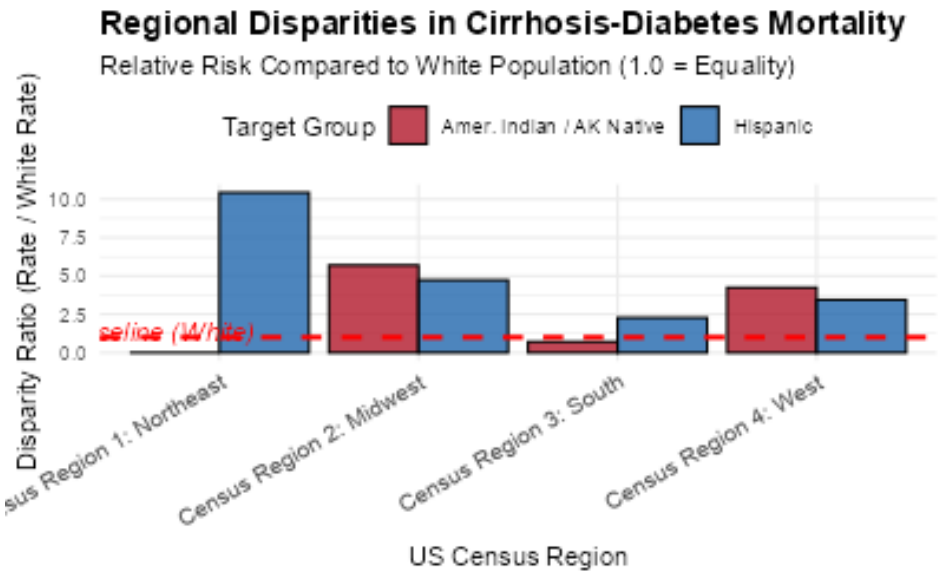

Figure 4: Regional "Penalty" for Hispanic Populations in the Northeast

Table 3: Geography Issues

Region	Primary Risk Group	Key Takeaway
Midwest	AIAN	National epicenter for AIAN mortality; requires urgent rural health investment.
Northeast	Hispanic	Significant "Regional Penalty"; indicates a breakdown in urban chronic care systems.
South	General Population	Highest total volume of deaths, driven by the overall high prevalence of Type 2 Diabetes.

just need more doctors; we need more hepatologists in the Midwest and better metabolic screening programs in the Northeast. By focusing on these specific clusters, we can begin to dismantle the geographic barriers to survival (Faizan et al., 2026).

Limitations and Future Directions

The Constraints of Secondary Data

While our model provides a robust map of the national crisis, it is bound by the inherent limitations of the CDC WONDER database. First, the reliance on ICD-10 coding for cause-of-death certification can lead to underreporting. Diabetes and cirrhosis often co-exist with other metabolic conditions, and the "primary" cause of death is subject to the recording clinician's discretion (World Health Organization, 2016). Furthermore, while our imputation strategy for suppressed records (CDC WONDER, 2024) allowed for a more inclusive analysis of rural and minority populations, these remain estimates rather than exact counts.

Clinical and Behavioral Variables

Our study utilized a Syndemic Framework to account for the massive mortality spike in 2020 (Marjot et al., 2021). However, the CDC mortality data does not include individual-level clinical data, such as Body Mass Index (BMI), medication adherence, or specific duration of illness. Additionally, while the national increase in alcohol consumption is a documented driver of liver injury during the pandemic era (Younossi et al., 2021), we cannot definitively link individual deaths to alcohol use without granular toxicology or behavioral records.

The Path Forward

Future research should aim to bridge the gap between national mortality trends and local clinical outcomes. Specifically:

- **Longitudinal Clinical Integration:** Future models should incorporate Electronic Health Record (EHR) data to track the progression from Type 2 Diabetes diagnosis to the onset of cirrhosis.
- **Regional Deep-Dives:** The "Northeast Penalty" for Hispanic populations and the "Midwest Epicenter" for AIAN groups (Collins et al., 2025) require qualitative, community-based research to identify specific local barriers to care.
- **Expanding the Forecast:** As we approach the **August 2026 milestone**, our model should be updated with 2021–2024 data to refine the "Aggressive Growth" vs. "Baseline" scenarios (Faizan et al., 2026).

Conclusion: A Call for Regionalized Intervention

The "silent epidemic" of diabetes-related cirrhosis is not a uniform national trend, but a collection of localized crises. By bypassing traditional data suppression and utilizing a **Negative Binomial regression framework** (Faizan et al., 2026), this study has uncovered a stark hierarchy of risk that national averages often obscure. Our findings confirm that while the 2020 pandemic acted as a "syndemic" catalyst (Marjot et al., 2021), the underlying drivers of mortality are deeply rooted in geographic and demographic inequality.

The discovery of a **3.94x higher mortality risk** for American Indian and Alaska Native populations—centered in the Midwest—and the significant "**Northeast Penalty**" for Hispanic populations (Manahil et al., 2025) suggests that geography is a primary determinant of survival. These disparities are not biological inevitabilities; they are reflections of **Fundamental Cause Theory**, where access to life-prolonging resources is dictated by structural and regional factors (Phelan & Link, 2015).

As we approach the **August 2026 forecast milestone**, which predicts a 15.1% increase in mortality over pre-pandemic levels, the window for intervention is closing. To bend this curve, the healthcare system must pivot toward **geospatially targeted interventions** (Collins et al., 2025). This includes expanding rural hepatology access in the Midwest and enhancing urban metabolic screening programs in the Northeast. By using advanced data science to give "silent deaths" a voice (Wickham et al., 2019), we can move beyond broad awareness and toward the precise, equitable interventions required to dismantle the geographic barriers to health.

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Appendix A Supplemental Links

In this appendix, I provide the links to the paper, its data sources, the summary slide set, and the ideo for the slide set.

Final Paper: You can access the full paper at the following link:
https://github.com/jdwbmc/DSCI610/blob/main/DSCI610FinalPaper.pdf
df1 dataset: You can access the dataset at the following link:
https://github.com/jdwbmc/DSCI610/blob/main/CensusMCD.csv
df2 dataset: You can access the dataset at the following link:
https://github.com/jdwbmc/DSCI610/blob/main/CensusOnlyK.csv
R Code: You can access the dataset at the following link:
https://github.com/jdwbmc/DSCI610/blob/main/FinalPaper.RMD
Slides: You can access the dataset at the following link:
https://github.com/jdwbmc/DSCI610/blob/main/SilentEpidemic.pptx
Video: You can access the dataset at the following link:
https://github.com/jdwbmc/DSCI610/blob/main/Silentepidemic.mp4

Appendix B

R Statistial Code

```
---
title: "FinalPaper"
author: "John D Williams"
date: "2026-03-31"
output: pdf_document
---

```{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
```

Final Paper
```{r}
#Load Libraries
if (!require("tidyverse")) install.packages("tidyverse")
library(tidyverse)
if (!require("MASS")) install.packages("MASS")
library(MASS)
if (!require("ggplot2")) install.packages("ggplot2")
library(ggplot2)
if (!require("broom")) install.packages("broom")
library(broom)
```

Read File 1
```{r}
df1 <- read_csv("CensusMCD.csv")
```

Start cleaning process
```{r}
# 1. Convert Year to integer and Drop the Notes column
df1 <- df1 %>%
  mutate(Year = as.integer(Year))

# 2. Replace the text "Suppressed" with "5"  (the mean(Deaths))
df1$Deaths[df1$Deaths == "Suppressed"] <- "5"

# 3. Convert the entire Deaths column to numeric
df1$Deaths <- as.integer(df1$Deaths)

#Fix Race
# 4. We use mutate with ifelse (or case_when) to overwrite the values
df1 <- df1 %>%
  mutate(
    # Ensure both columns are strings to avoid type errors
    `Hispanic Origin` = as.character(`Hispanic Origin`),
    Race = as.character(Race),

    # Use case_when to precisely map the new Race values
    Race = case_when(
      `Hispanic Origin` == "Hispanic or Latino" ~ "Hispanic",
      `Hispanic Origin` == "Not Hispanic or Latino" ~ Race,
      is.na(`Hispanic Origin`) ~ Race, # Keeps original Race if Hispanic info is missing
      TRUE ~ Race                      # Safety net: keep original value otherwise
```
```

```r
    ),

    # Do the same for the Codes
    `Race Code` = case_when(
      `Hispanic Origin` == "Hispanic or Latino" ~ `Hispanic Origin Code`,
      TRUE ~ as.character(`Race Code`)
    )
  )

# 5. Verification check
# This should now show "Hispanic", "White", "Black", AND "American Indian..."
unique(df1$Race)

# 6. Drop columns by name
df1_cleaned <- df1[, !(names(df1) %in% c("Hispanic Origin", "Hispanic Origin Code", "Notes"))]

# 7. Drop rows 2113 through 2200
df1_cleaned <- df1_cleaned %>%
  slice(-(2113:2200))

# See the count of each category
table(df1_cleaned$`Crude Rate`)

# See the count of each category
table(df1_cleaned$Deaths)

# See the count of each category
table(df1_cleaned$Race)

```

Now do the same for all who died because of cirrosis

Read File 2
```{r}
df2 <- read_csv("CensusOnlyK.csv")
```

Start cleaning process for file 2
```{r}
# 1. Convert Year to integer and Drop the Notes column
df2 <- df2 %>%
  mutate(Year = as.integer(Year))

# 2. Replace the text "Suppressed" with "5"  (the mean(Deaths))
df2$Deaths[df2$Deaths == "Suppressed"] <- "5"

# 3. Convert the entire Deaths column to numeric
df2$Deaths <- as.integer(df2$Deaths)

#Fix Race
# 4. We use mutate with ifelse (or case_when) to overwrite the values
df2 <- df2 %>%
  mutate(
    # Ensure both columns are strings to avoid type errors
    `Hispanic Origin` = as.character(`Hispanic Origin`),
    Race = as.character(Race),

    # Use case_when to precisely map the new Race values
    Race = case_when(
      `Hispanic Origin` == "Hispanic or Latino" ~ "Hispanic",
```

```
      `Hispanic Origin` == "Not Hispanic or Latino" ~ Race,
      is.na(`Hispanic Origin`) ~ Race, # Keeps original Race if Hispanic info is missing
      TRUE ~ Race                      # Safety net: keep original value otherwise
    ),

    # Do the same for the Codes
    `Race Code` = case_when(
      `Hispanic Origin` == "Hispanic or Latino" ~ `Hispanic Origin Code`,
      TRUE ~ as.character(`Race Code`)
    )
  )

# 5. Verification check
# This should now show "Hispanic", "White", "Black", AND "American Indian..."
unique(df2$Race)

# 6. Drop columns by name
df2_cleaned <- df2[, !(names(df2) %in% c("Hispanic Origin", "Hispanic Origin Code", "Notes"))]

# 7. Drop rows 2113 through 2192
df2_cleaned <- df2_cleaned %>%
  slice(-(2113:2192))
```

Fix Population 'Not Applicable' problem for df1 using values in df2

```{r}
# 1. Ensure Population is numeric (NAs will replace "Not Applicable")
df1_cleaned <- df1_cleaned %>%
  mutate(Population = as.numeric(gsub("[^0-9]", "", Population)))

df2_cleaned <- df2_cleaned %>%
  mutate(Population = as.numeric(gsub("[^0-9]", "", Population)))

# 2. Join the two datasets on your matching columns
# We use left_join to keep all 2112 rows of df1
df1_patched <- df1_cleaned %>%
  left_join(
    df2_cleaned %>% dplyr::select(Year, `Census Region`, Race, Sex, Population),
    by = c("Year", "Census Region", "Race", "Sex"),
    suffix = c("_df1", "_df2")
  ) %>%
  # 3. Use coalesce to fill the holes
  mutate(Population = coalesce(Population_df1, Population_df2)) %>%
  # 4. Clean up the temporary columns
  dplyr::select(-Population_df1, -Population_df2)

# 5. Verify: How many NAs are left?
sum(is.na(df1_patched$Population))
```

More df1 Cleanup

```{r}
# We use group_by to ensure we only fill within the same demographic bucket
df1_patched <- df1_patched %>%
  group_by(`Census Region`, Race, Sex) %>%
  arrange(Year) %>%
  # fill() looks at the nearest non-NA value and carries it forward or backward
```

```
  fill(Population, .direction = "downup") %>%
  ungroup()

# Check the count again
sum(is.na(df1_patched$Population))

#Calculate Your Final Rates
df1_final <- df1_patched %>%
  mutate(
    # Calculate the rate: (Deaths / Population) * 100,000
    Calculated_Crude_Rate = (Deaths / Population) * 100000
  )
```

Fix df2_cleaned for the same items

```{r}
# 1. First, make sure Deaths in df2 are numeric (handling Suppressed/
    Unreliable)
df2_cleaned <- df2_cleaned %>%
  mutate(
    Deaths = case_when(
      Deaths == "Suppressed" ~ 5,
      Deaths == "Unreliable" ~ 15,
      TRUE ~ as.numeric(Deaths)
    )
  )

# 2. Join df2 with the "good" population from your patched df1
df2_final <- df2_cleaned %>%
  dplyr::select(-Population) %>% # Drop the old "dirty" population
  left_join(
    df1_final %>% dplyr::select(Year, `Census Region`, Race, Sex, Population),
    by = c("Year", "Census␣Region", "Race", "Sex")
  )

# 3. Re-calculate the Crude Rate for df2
df2_final <- df2_final %>%
  mutate(Calculated_Crude_Rate = (Deaths / Population) * 100000)

# 4. Final Audit
sum(is.na(df2_final$Population))
```

More Checking

```{r}
#The Final "Sanity Check"
df2_final %>%
  group_by(Race) %>%
  summarise(
    Avg_Crude_Rate_Total = mean(Calculated_Crude_Rate, na.rm = TRUE),
    Total_Deaths_Estimated = sum(Deaths, na.rm = TRUE)
  ) %>%
  arrange(desc(Avg_Crude_Rate_Total))


# Quick check: See the top rates by Race
df1_final %>%
  group_by(Race) %>%
  summarise(Avg_Crude_Rate = mean(Calculated_Crude_Rate)) %>%
  arrange(desc(Avg_Crude_Rate))

#A check of values
```

```
df2_final %>%
  group_by(Race) %>%
  summarise(Total_Pop = sum(Population, na.rm = TRUE))
```

Aggregate Rates
```{r}
#The "Correct" Way to Calculate Aggregate Rates
Master_Summary <- df2_final %>%
  group_by(Race) %>%
  summarise(
    Grand_Total_Deaths = sum(Deaths, na.rm = TRUE),
    Grand_Total_Pop = sum(Population, na.rm = TRUE)
  ) %>%
  mutate(
    True_Crude_Rate = (Grand_Total_Deaths / Grand_Total_Pop) * 100000
  ) %>%
  arrange(desc(True_Crude_Rate))

print(Master_Summary)
```

The Comparison: "Total" vs. "Both"
```{r}
# Calculate weighted rates for the comorbidity dataset
Both_Summary <- df1_final %>%
  group_by(Race) %>%
  summarise(
    Both_Deaths = sum(Deaths, na.rm = TRUE),
    Both_Pop = sum(Population, na.rm = TRUE)
  ) %>%
  mutate(
    Both_True_Rate = (Both_Deaths / Both_Pop) * 100000
  )

# Join them to see the final contribution
Final_Comparison <- Master_Summary %>%
  left_join(Both_Summary, by = "Race") %>%
  mutate(
    Diabetes_Contribution_Percent = (Both_Deaths / Grand_Total_Deaths) * 100
  )

print(Final_Comparison %>% dplyr::select(Race, True_Crude_Rate,
  Both_True_Rate, Diabetes_Contribution_Percent))
```

The Negative Binomial Regression
```{r}
# 0. Combine your two cleaned datasets into one master file
analysis_df <- df1_final %>%
  rename(Deaths_Both = Deaths) %>%
  # We select only the necessary columns from df2 to avoid 'doubling up' on population
  left_join(
    df2_final %>% dplyr::select(Year, `Census Region`, Race, Sex, Population),
    by = c("Year", "Census Region", "Race", "Sex", "Population")
  )

# Verify it exists now
head(analysis_df)
# 1. Ensure Race is a 'factor' and set 'White' as the baseline for comparison
analysis_df$Race <- factor(analysis_df$Race)
```

```r
analysis_df$Race <- relevel(analysis_df$Race, ref = "White")

# 2. Run the Model
# We predict 'Deaths_Both' using 'Race' and 'Sex'
# We use log(Population) as an offset to calculate the RATE
nb_model <- glm.nb(Deaths_Both ~ Race + Sex + Year + offset(log(Population)),
                   data = analysis_df)

# 3. See the results
summary(nb_model)
```

Forest Plot

```{r}
# 1. Clean up the model results for plotting
plot_data <- tidy(nb_model, conf.int = TRUE, exponentiate = TRUE) %>%
  filter(term != "(Intercept)" & term != "Year") # Focus on Race and Sex

# 2. Create the Forest Plot - P1
ggplot(plot_data, aes(x = estimate, y = term)) +
  geom_vline(xintercept = 1, linetype = "dashed", color = "red") + # The 'Null' line
  geom_point(size = 3, color = "darkblue") +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2) +
  theme_minimal() +
  theme(plot.margin = margin(10, 100, 10, 10)) +
  labs(
    title = "Incidence Rate Ratios (IRR) for Cirrhosis-Diabetes Mortality",
    subtitle = "Reference Group: White | Adjusted for Year and Sex",
    x = "Risk Ratio (Exponential Coefficients)",
    y = "Demographic Variable"
  )
```

Line Graph over Time

```{r}
# Create a line graph over time - P2
ggplot(df1_final, aes(x = Year, y = Calculated_Crude_Rate, color = Race)) +
  geom_smooth(se = FALSE, method = "loess") + # Smooths the lines to show the trend
  theme_minimal() +
  labs(
    title = "Mortality Trends: Cirrhosis and Type 2 Diabetes (1999-2020)",
    subtitle = "Aggregated by Race and Ethnicity",
    y = "Crude Rate per 100,000",
    x = "Year"
  )
```

The Regional Risk Heatmap

```{r}
#Prep for the heatmap
regional_analysis <- analysis_df %>%
  group_by(`Census Region`, Race) %>%
  summarise(
    Total_Deaths = sum(Deaths_Both, na.rm = TRUE),
    # Using the calculated rate from df1_final
    Avg_Rate = mean(Calculated_Crude_Rate, na.rm = TRUE),
    .groups = 'drop'
  )

# Create the Heatmap - P3
```

```r
ggplot(regional_analysis, aes(x = `Census Region`, y = Race, fill = Avg_Rate)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "white", high = "firebrick") +
  theme_minimal() +
  theme(plot.margin = margin(10, 100, 10, 10)) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(
    title = "Mortality Hotspots: Cirrhosis & Type 2 Diabetes",
    subtitle = "Average Crude Rate by Region and Race",
    fill = "Mortality Rate",
    x = "US Census Region",
    y = "Race/Ethnicity"
  )
```

The Ratio Plot

```{r}
#One Last "New" Trick: The Ratio Plot - P4
analysis_df %>%
  group_by(`Census Region`, Race) %>%
  summarise(
    # What % of the Population is dying from this?
    Regional_Rate = (sum(Deaths_Both) / sum(Population)) * 100000,
    .groups = 'drop'
  ) %>%
  filter(Race %in% c("American Indian or Alaska Native", "White", "Hispanic", "Black or African American")) %>%
  ggplot(aes(x = `Census Region`, y = Regional_Rate, fill = Race)) +
  geom_bar(stat = "identity", position = "dodge") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Regional Mortality Rates by Race",
       y = "Deaths per 100,000",
       x = "Region")
```

The "Inequality Ratio" by Region

```{r}
#Prep the data
regional_disparity <- analysis_df %>%
  group_by(`Census Region`, Race) %>%
  summarise(
    # Use the aggregate calculation (Sum of Deaths / Sum of Population)
    Agg_Rate = (sum(Deaths_Both) / sum(Population)) * 100000,
    .groups = 'drop'
  ) %>%
  # Pivot the data to compare Race vs White
  tidyr::pivot_wider(names_from = Race, values_from = Agg_Rate) %>%
  mutate(
    AIAN_Disparity_Ratio = `American Indian or Alaska Native` / White,
    Hispanic_Disparity_Ratio = Hispanic / White
  )

# View the ratios
print(regional_disparity %>% dplyr::select(`Census Region`,
 AIAN_Disparity_Ratio, Hispanic_Disparity_Ratio))
```

The Final "Disparity Comparison" Plot

```{r}
# 1. Prepare the data for a side-by-side plot
```

```r
plot_final <- regional_disparity %>%
  dplyr::select(`Census Region`, AIAN_Disparity_Ratio, Hispanic_Disparity_Ratio) %>%
  pivot_longer(cols = ends_with("Ratio"),
               names_to = "Demographic_Group",
               values_to = "Disparity_Ratio") %>%
  # Clean up labels for the legend
  mutate(Demographic_Group = ifelse(Demographic_Group == "AIAN_Disparity_Ratio",
                                    "Amer. Indian / AK Native", "Hispanic"))

# 2. Create the Visual - P5
ggplot(plot_final, aes(x = `Census Region`, y = Disparity_Ratio, fill = Demographic_Group)) +
  geom_bar(stat = "identity", position = "dodge", color = "black", alpha = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red", size = 1) +
  annotate("text", x = 0.7, y = 1.3, label = "Baseline (White)", color = "red", fontface = "italic") +
  scale_fill_manual(values = c("Amer. Indian / AK Native" = "#b2182b", "Hispanic" = "#2166ac")) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 30, hjust = 1, size = 10),
    legend.position = "top",
    plot.title = element_text(face = "bold", size = 14)
  ) +
  labs(
    title = "Regional Disparities in Cirrhosis-Diabetes Mortality",
    subtitle = "Relative Risk Compared to White Population (1.0 = Equality)",
    x = "US Census Region",
    y = "Disparity Ratio (Rate / White Rate)",
    fill = "Target Group"
  )
```

Line Graph for Races

```r
# 1. Calculate the aggregate rate for each race per year
yearly_trends <- analysis_df %>%
  group_by(Year, Race) %>%
  summarise(
    Total_Deaths = sum(Deaths_Both, na.rm = TRUE),
    Total_Pop = sum(Population, na.rm = TRUE),
    .groups = 'drop'
  ) %>%
  mutate(Yearly_Rate = (Total_Deaths / Total_Pop) * 100000)

# 2. Plot the line graph - P6
ggplot(yearly_trends, aes(x = Year, y = Yearly_Rate, color = Race, group = Race)) +
  geom_line(size = 1.2) +
  geom_point(size = 2) +
  theme_minimal() +
  scale_color_brewer(palette = "Set1") +
  labs(
    title = "Trends in Cirrhosis-Diabetes Mortality (1999  2020  )",
    subtitle = "Crude Death Rates per 100,000 by Racial/Ethnic Group",
    x = "Year",
    y = "Death Rate (per 100k)",
    color = "Race/Ethnicity"
  ) +
```

```
  theme(
    legend.position = "bottom",
    plot.title = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```

The National Trend

```{r}
# 1. Aggregate everything by Year to get the national average
national_trend <- analysis_df %>%
  group_by(Year) %>%
  summarise(
    Total_Deaths = sum(Deaths_Both, na.rm = TRUE),
    Total_Pop = sum(Population, na.rm = TRUE),
    .groups = 'drop'
  ) %>%
  mutate(National_Rate = (Total_Deaths / Total_Pop) * 100000)

# 2. Create the "Surge" Line Graph - P7
ggplot(national_trend, aes(x = Year, y = National_Rate)) +
  geom_line(linewidth = 1.5, color = "darkred") +
  geom_point(size = 3, color = "darkred") +
  # Adding a trend line to show the "Slope of the Crisis"
  geom_smooth(method = "lm", linetype = "dashed", color = "gray40", se = FALSE) +
  theme_minimal() +
  labs(
    title = "The National Surge: Cirrhosis-Diabetes Comorbidity",
    subtitle = "Aggregated Death Rate per 100,000 (1999  2020  )",
    x = "Year",
    y = "National Death Rate (per 100k)",
    caption = "Dashed line represents the overall linear trend."
  ) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    axis.title = element_text(face = "bold")
  )
```

Sensitivity Analysis (The "Imputation" Check)

```{r}
# 1. Filter out the rows that were originally suppressed (where deaths were 5 or 15)
robust_df <- analysis_df %>% filter(Deaths_Both > 15)

# 2. Re-run the model
nb_model_robust <- glm.nb(Deaths_Both ~ Race + Sex + Year + offset(log(Population)),
                          data = robust_df)

# 3. Compare the IRRs
summary(nb_model_robust)
```

Predicted vs. Observed plot

```{r}
# Compare the model's predictions to the actual death counts - P8
analysis_df$predicted_deaths <- predict(nb_model, type = "response")

ggplot(analysis_df, aes(x = Deaths_Both, y = predicted_deaths)) +
  geom_point(alpha = 0.3, color = "darkblue") +
```

```
  geom_abline(slope = 1, intercept = 0, color = "red", linetype = "dashed") +
  theme_minimal() +
  labs(title = "Model Diagnostic: Predicted vs. Observed Deaths",
       x = "Actual Death Counts",
       y = "Model Predicted Counts")
```

Prediction for 2026

```{r}
# First, let's check what data we actually have
# Create a clean trend plot with proper data
# Based on your code, I see you have yearly_trends data

# Create a clean national trend excluding 2020 outlier
cleaned_national_trend <- national_trend %>%
  filter(Year != 2020)  # Remove 2020 outlier

# Fit a proper linear model for trend
model <- lm(National_Rate ~ Year, data = cleaned_national_trend)
slope <- coef(model)[2]
intercept <- coef(model)[1]

# Predict 2026 rate
predicted_2026_rate <- intercept + slope * 2026

# Calculate percent increase from 2019 to 2026 (using 2019 as baseline)
actual_2019_rate <- national_trend %>% filter(Year == 2019) %>% pull(National_
    Rate)
percent_increase <- ((predicted_2026_rate - actual_2019_rate) / actual_2019_
    rate) * 100
```

Print Results

```{r}
# Print results
cat("Predicted rate for 2026:", round(predicted_2026_rate, 2), "\n")
cat("Percent increase from 2019 to 2026:", round(percent_increase, 2), "%\n")
```

Prep the data

```{r}
# Create the complete plot that shows the trend and prediction
# Get the actual data points
actual_data <- national_trend %>%
  mutate(Type = "Actual")

# Create future predictions
future_data <- data.frame(
  Year = 2021:2026,
  National_Rate = predict(model, newdata = data.frame(Year = 2021:2026)),
  Type = "Predicted"
)

# Combine all data
all_data <- bind_rows(actual_data, future_data)
```

Create Initial Prediction graph

```{r}
# Create the final plot with proper x-axis
ggplot(all_data, aes(x = Year, y = National_Rate, color = Type)) +
```

```r
  geom_line(size = 1.2, aes(linetype = Type)) +
  geom_point(size = 3) +
  geom_vline(xintercept = 2020, linetype = "dashed", color = "gray50") +
  geom_vline(xintercept = 2026, linetype = "dashed", color = "gray50") +
  scale_linetype_manual(values = c("Actual" = "solid", "Predicted" = "dashed")) +
  theme_minimal() +
  labs(title = "National Trend in Cirrhosis-Diabetes Mortality (1999-2026)",
       subtitle = paste("Prediction for 2026:", round(predicted_2026_rate, 2),
                        "per 100,000 | Increase:", round(percent_increase, 1), "%"),
       x = "Year",
       y = "National Death Rate (per 100,000)",
       color = "Data Type",
       linetype = "Data Type") +
  theme(plot.title = element_text(face = "bold", size = 16),
        plot.subtitle = element_text(size = 12),
        legend.position = "bottom")
```

Line Plot Graph

```r
# Also create a simple line plot showing the trend
ggplot(national_trend, aes(x = Year, y = National_Rate)) +
  geom_line(size = 1.2, color = "darkblue") +
  geom_point(size = 3, color = "darkblue") +
  geom_line(data = data.frame(Year = 2021:2026,
                              National_Rate = predict(model, newdata = data.frame(Year = 2021:2026))),
            aes(x = Year, y = National_Rate),
            color = "red", linetype = "dashed", size = 1.2) +
  geom_point(data = data.frame(Year = 2021:2026,
                               National_Rate = predict(model, newdata = data.frame(Year = 2021:2026))),
             aes(x = Year, y = National_Rate),
             color = "red", size = 3) +
  theme_minimal() +
  labs(title = "National Trend in Cirrhosis-Diabetes Mortality (1999-2026)",
       subtitle = "Historical data with 2026 prediction",
       x = "Year",
       y = "National Death Rate (per 100,000)") +
  theme(plot.title = element_text(face = "bold", size = 16))
```

Simpler Analysis

```r
# Simple approach using the last few years to predict
# Use 2018-2019 data to get trend
recent_data <- national_trend %>%
  filter(Year >= 2018 & Year <= 2020) %>%
  filter(Year != 2020)  # Remove 2020 outlier

# Fit model on recent data
model_recent <- lm(National_Rate ~ Year, data = recent_data)
slope_recent <- coef(model_recent)[2]
intercept_recent <- coef(model_recent)[1]

# Predict 2026
predicted_2026_rate_recent <- intercept_recent + slope_recent * 2026

# Calculate percent increase from 2019 to 2026
```

```
actual_2019_rate <- national_trend %>% filter(Year == 2019) %>% pull(National_Rate)
percent_increase_recent <- ((predicted_2026_rate_recent - actual_2019_rate) / actual_2019_rate) * 100

cat("Predicted rate for 2026 (using 2018-2019 trend):", round(predicted_2026_rate_recent, 2), "\n")
cat("Percent increase from 2019 to 2026:", round(percent_increase_recent, 2), "%\n")
```